Project Title: 2D Scene with Movement and Rotation Controls for Shapes

Description

This project is a simple 2D OpenGL scene that features six distinct shapes arranged in a circular layout. Each shape (triangle, rectangle, circle, pentagon, line, and oval) has unique movement and rotation controls, allowing the user to manipulate them independently using the keyboard. This project demonstrates transformations in OpenGL, including translation and rotation, applied individually to multiple objects.

Instructions

Keyboard Controls

- **Shape Movements and Rotation** Each shape has a unique set of movement keys assigned to control its position. Every time a movement key is pressed, the shape will also rotate by a specified angle (10 degrees by default).

Shape	Up	Down	Left	Right
Triangle	w	s	a	d
Rectangle	r	f	e	t
Circle	y	h	g	j
Pentagon	i	k	u	o
Line	p	;	l	[
Oval	z	x	c	v

- **Reset Positions and Rotation**
  - `q` - Resets all shapes to their original positions and rotations.
- **Exit**
  - `Esc` - Closes the program.

Mouse Controls

- **No mouse controls** are implemented in this project. All interactions are managed through the keyboard.

Creative Aspects

1. **Independent Rotation and Movement for Each Shape**: Each shape has its own unique set of keys for movement and rotation. This provides the user with a more interactive and controlled experience in manipulating the shapes on the screen.
2. **Circular Layout with Defined Colors**: The initial layout of the shapes in a circular pattern provides an aesthetic arrangement, giving the user a sense of organization and balance when starting the application.
3. **Real-Time Transformations**: The project demonstrates real-time transformations, where each key press updates both the position and rotation angle of each shape, showcasing fundamental graphics programming techniques.